Wendian **Luo**

Sichuan Unversity Wangjiang Campus, Wuhou District, Chengdu, China, 610065

+86 191 8096 4043 | 1850245174@qq.com | mr-lwd.github.io/wendianluo.github.io/ | Mr-lwd

Research interests: Edge Intelligence, Federated Learning, Computer Vision, Large Language Models, and Embodied Intelligence

Education

Sichuan University, College of Computer Science — *Chengdu, China*

M.Eng. in Computer Technology — *Sep. 2024 – Jun. 2027 (Expected)*

- GPA: 3.65/4.00.
- First-Class Graduate Academic Scholarship.

Sichuan University, College of Computer Science — *Chengdu, China*

B.Eng. in Computer Science and Technology — *Sep. 2020 – Jun. 2024*

- GPA: 3.72/4.00; GPA ranking: 38/318 (top 12%); graduate recommendation ranking: 8/318.
- Outstanding Graduate of Sichuan Province, 2024.

Research Experience

TMSC: Zero-Shot Industrial Anomaly Detection — *Open-Source Research Project*

Researcher and Developer · Project Repository — *2026*

- Developed a zero-shot anomaly detection pipeline using DINOv3 and CLIP, integrating multi-layer representations, dual-branch prompt adapters, cross-modal attention, and three-region semantic calibration.

FedSAE: Semi-Asynchronous Energy-Efficient Federated Prototype Learning — *Sichuan University*

First Author and Researcher — *2025*

- Designed a semi-asynchronous end–edge–cloud federated prototype learning framework that exchanges class-level feature statistics instead of raw data or full model parameters.
- Validated the energy-saving effectiveness of the framework's dynamic DVFS mechanism through experiments on an NVIDIA Jetson TX2 edge platform.

LPSAFS: Low-Power Scheduling with Segmented Frequency Scaling — *Sichuan University*

First Author and Researcher — *2024*

- Designed a segmented frequency-scaling algorithm based on minimum-slack allocation for periodic real-time tasks under discrete DVFS states, deadline constraints, and reliability requirements.

Publications

Journal Articles

1. **Wendian Luo**, Tong Yu, Junnan Li, Shengxin Dai, Bing Guo, and Zhuohang Jiang, "TMSC: A Tri-Regional Mask-Guided Semantic Calibration Framework for Zero-Shot Anomaly Detection," ***Pattern Recognition*** (Minor Revision).
2. **Wendian Luo**, Tong Yu, Shengxin Dai, Bing Guo, Xuesen Lin, and Yanglin Pu, "Semi-Asynchronous Energy-Efficient Federated Prototype Learning for End-Edge-Cloud Architectures," ***Future Generation Computer Systems***, vol. 179, Art. no. 108351, 2026. https://doi.org/10.1016/j.future.2025.108351
3. **Wendian Luo**, Shengxin Dai, Cheng Dai, Bing Guo, Sherif Moussa, and Mubarak Alrashoud, "BDTest: A Diversity-Oriented Test Case Generation Framework for Deep Neural Networks in 6G-IOT," ***IEEE Internet of Things Journal***, vol. 13, no. 5, pp. 8190–8203, 2026. https://doi.org/10.1109/JIOT.2025.3616358
4. Shengxin Dai, Zheng Luo, **Wendian Luo**, Siyi Wang, Cheng Dai, Bing Guo, and Xiaokang Zhou, "Energy-Efficient Inference With Software–Hardware Co-Design for Sustainable Artificial Intelligence of Things," ***IEEE Internet of Things Journal***, vol. 11, no. 24, pp. 39170–39182, 2024. https://doi.org/10.1109/JIOT.2024.3482288

Conference Papers

1. **Wendian Luo**, Shengxin Dai, and Bing Guo, "A Low Power Scheduling Algorithm for Segmented Frequency Scaling," in ***2025 IEEE International Symposium on Parallel and Distributed Processing with Applications (ISPA)***, pp. 828–837, 2025. https://doi.org/10.1109/ISPA67752.2025.00112

Professional Experience

Media Intelligence Laboratory, Chengdu Sobey — *Chengdu, China*

Large Language Model Algorithm R&D Intern — *Jul. 2024 – Sep. 2024*

- Constructed and cleaned English instruction-tuning datasets for Story-to-Scene generation, scene keyword extraction, and shot-level visual description tasks; developed scripts for translation, provenance tracing, deduplication, and format validation.
- Fine-tuned and evaluated Qwen-72B with LLaMA-Factory and LoRA on an eight-GPU NVIDIA A800 server.
- Deployed inference services using vLLM, Flask, and HTTP APIs, and completed capability integration, API documentation, and project handover.

Selected Projects

Remote Relay: Cross-Device Codex Workspace — *React · FastAPI · Codex CLI*

Project Developer · Open Source — *GitHub: remote-codex-relay*

- Built a self-hosted React + FastAPI web workspace for the local Codex CLI, enabling session access across computers, tablets, and phones with live task updates, approval handling, and file exchange.
- Implemented one-command startup, dependency installation, frontend builds, and directory migration repair, preserving existing configuration and session data.

OpenClaw-Enabled Mobile Robot Control and Vision-Guided Grasping — *ROSMASTER M3 PRO*

Project Developer · System Integration and Deployment — *May 2026 – Jul. 2026*

- Deployed and configured an OpenClaw-based natural-language robot-control platform integrating MCP/FastMCP with ROS 2, enabling tool-based access to the omnidirectional chassis, 6-DoF arm, and perception modules.
- Integrated and validated a closed-loop RGB-D tracking-and-grasping workflow spanning visual grounding, mobile-base alignment, 3D grasp-point estimation, inverse-kinematics control, and collision-aware retreat; tuned grasp offsets and diagnosed visual/depth localization errors.

PaddlePaddle-Based 3D Medical Data Analysis Platform — *12th China Software Cup*

Primary Developer · National Second Prize — *Jun. 2023 – Aug. 2023*

- Developed the medical platform, 3D imaging workspace, and UI/UX using Vue.js, Vuetify, and VTK.js; contributed to the Python/Django backend and Tencent Cloud COS integration.
- Integrated PaddlePaddle MedicalSeg into an end-to-end workflow covering medical-image upload, AI-assisted segmentation, 3D visualization, and patient management.

Tingyin: Deep Learning-Based Acoustic Life Detection for Earthquake Rescue — *Sichuan University*

Core Team Member · National Innovation Training Program — *Oct. 2022 – Sep. 2023*

- Developed the mobile application, conducted hardware feasibility analysis, and contributed to algorithm deployment and end-to-end system integration.
- Integrated TSTNN-based speech enhancement and Whisper-based speech recognition; completed the project ahead of schedule and received First Prize in the Sichuan Division of the 2023 Chinese College Students Computer Design Competition.

Technical Skills

Programming	Python, C, LaTeX
Machine Learning	PyTorch, CLIP, DINOv3, federated learning, zero-shot anomaly detection
Edge AI	NVIDIA Jetson Thor/AGX/Orin, DVFS, Model deployment, vLLM, Ollama
AI Development	Cursor, Codex, Trae, OpenClaw
Languages	Chinese (native); English (CET-4 and CET-6); Prepare for IELTS

Honors & Awards

2024	**Outstanding Graduate of Sichuan Province**, Sichuan Province	*China*
2024–27	**First-Class Graduate Academic Scholarship**, Sichuan University	*Chengdu*
2023	**National Second Prize**, 12th China Software Cup	*China*
2023	**First Prize, Sichuan Division**, Chinese College Students Computer Design Competition	*Sichuan*
2020–24	**Merit and Academic Scholarships**, Sichuan University	*Chengdu*